

## VINJEORA II, Norway

WMO: 012320

Lat: 63.200N

Lon: 9.000E

Elev: 220

StdP: 14.58

Time Zone: 1.00 (EUC)

Period: 10-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |     |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|-----|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99% |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP  | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g) | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | 11.0       | 14.3 | -1.2                          | 5.2 | 13.8 | 1.8 | 6.1 | 17.5 | 22.6                  | 36.5 | 20.7 | 34.5 | 9.6                   | 170  | 0.596 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month<br>DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD<br>to 0.4% DB |      |
|---------------|---------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|-------------------------|------|
|               |                           | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                         |      |
|               |                           | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                    | PCWD |
| (a)           | (b)                       | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                     | (p)  |
| 7             | 14.8                      | 76.2            | 60.6 | 72.7 | 58.8 | 69.8 | 57.6 | 63.4                | 72.3 | 61.2 | 69.3 | 59.4 | 66.5 | 6.7                     | 160  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 59.6                            | 76.7 | 67.0 | 57.8 | 71.9 | 64.6 | 56.1 | 67.5 | 62.4 | 28.9          | 72.1 | 27.3 | 69.7 | 26.1 | 66.4 | 71.6              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |     |                    |     | n-Year Return Period Values of Extreme Temperature |     |            |     |            |     |            |      |      |     |
|-------------------|------|------|----------------------------|-----|--------------------|-----|----------------------------------------------------|-----|------------|-----|------------|-----|------------|------|------|-----|
|                   |      |      | Mean                       |     | Standard Deviation |     | n=5 years                                          |     | n=10 years |     | n=20 years |     | n=50 years |      |      |     |
| 1%                | 2.5% | 5%   | Min                        | Max | Min                | Max | Min                                                | Max | Min        | Max | Min        | Max | Min        | Max  |      |     |
| (a)               | (b)  | (c)  | (e)                        | (f) | (g)                | (h) | (i)                                                | (j) | (k)        | (l) | (m)        | (n) | (o)        | (p)  |      |     |
| 17.9              | 15.4 | 13.6 | DB                         | 9.3 | 83.4               | 3.1 | 4.6                                                | 7.0 | 86.8       | 5.2 | 89.5       | 3.4 | 92.1       | 1.2  | 95.4 | (4) |
|                   |      |      | WB                         | 7.6 | 66.4               | 3.3 | 2.6                                                | 5.2 | 68.3       | 3.2 | 69.8       | 1.4 | 71.3       | -1.1 | 73.2 | (5) |

## Monthly Climatic Design Conditions

|                                                                                  |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct  | Nov   | Dec  |
|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|-------|------|
|                                                                                  |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)  | (o)   | (p)  |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 44.4   | 30.3  | 32.8  | 35.6  | 43.0  | 50.5  | 54.3  | 59.3  | 58.1  | 52.8  | 43.8 | 37.8  | 33.7 |
|                                                                                  | DBStd       | 12.15  | 8.52  | 8.01  | 6.80  | 6.81  | 7.43  | 5.01  | 5.93  | 4.98  | 5.90  | 7.22 | 7.83  | 8.10 |
|                                                                                  | HDD50       | 3002   | 610   | 482   | 448   | 228   | 85    | 16    | 2     | 1     | 37    | 216  | 373   | 504  |
|                                                                                  | HDD65       | 7552   | 1075  | 902   | 913   | 660   | 451   | 321   | 197   | 222   | 368   | 657  | 817   | 969  |
|                                                                                  | CDD50       | 956    | 0     | 0     | 1     | 17    | 101   | 145   | 290   | 253   | 120   | 23   | 6     | 0    |
|                                                                                  | CDD65       | 32     | 0     | 0     | 0     | 0     | 2     | 1     | 20    | 8     | 1     | 0    | 0     | 0    |
|                                                                                  | CDH74       | 233    | 0     | 0     | 0     | 0     | 22    | 13    | 161   | 37    | 0     | 0    | 0     | 0    |
|                                                                                  | CDH80       | 39     | 0     | 0     | 0     | 0     | 2     | 0     | 35    | 2     | 0     | 0    | 0     | 0    |
| Wind                                                                             | WSAvg       | 6.0    | 8.5   | 7.7   | 6.5   | 5.6   | 5.4   | 4.5   | 4.1   | 4.1   | 4.6   | 6.2  | 6.9   | 7.6  |
| Precipitation                                                                    | PrecAvg     | 58.9   | 5.7   | 5.4   | 5.6   | 3.1   | 3.1   | 4.3   | 4.0   | 5.2   | 5.7   | 5.4  | 5.2   | 6.1  |
|                                                                                  | PrecMax     | 74.4   | 16.0  | 12.3  | 14.0  | 8.7   | 6.5   | 8.2   | 6.7   | 11.6  | 16.2  | 10.7 | 14.0  | 13.5 |
|                                                                                  | PrecMin     | 39.5   | 0.2   | 0.1   | 1.4   | 0.7   | 1.8   | 1.7   | 1.2   | 1.8   | 2.0   | 1.1  | 0.2   | 1.4  |
|                                                                                  | PrecStd     | 8.5    | 3.7   | 2.8   | 3.4   | 2.0   | 1.1   | 1.7   | 1.5   | 2.4   | 3.4   | 2.0  | 3.9   | 2.8  |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 50.1  | 54.9  | 56.7  | 66.1  | 76.4  | 75.4  | 84.7  | 78.3  | 71.1  | 65.7 | 61.1  | 54.1 |
|                                                                                  |             | MCWB   | 41.0  | 43.3  | 44.5  | 50.0  | 57.7  | 58.6  | 66.5  | 58.9  | 57.1  | 52.1 | 49.3  | 43.6 |
|                                                                                  | 2%          | DB     | 47.6  | 49.2  | 51.1  | 62.1  | 72.0  | 70.5  | 78.2  | 73.5  | 67.4  | 61.4 | 56.2  | 50.4 |
|                                                                                  |             | MCWB   | 38.9  | 39.5  | 41.0  | 48.0  | 55.7  | 56.3  | 63.1  | 58.4  | 55.6  | 50.5 | 46.5  | 40.7 |
|                                                                                  | 5%          | DB     | 44.9  | 46.7  | 48.2  | 57.9  | 68.3  | 67.4  | 74.1  | 70.3  | 64.8  | 57.6 | 52.2  | 47.3 |
|                                                                                  |             | MCWB   | 37.1  | 37.7  | 39.4  | 46.0  | 53.6  | 55.4  | 61.1  | 57.9  | 53.9  | 47.4 | 43.2  | 39.2 |
|                                                                                  | 10%         | DB     | 41.1  | 43.9  | 45.5  | 53.9  | 63.7  | 64.5  | 70.4  | 67.5  | 62.7  | 54.6 | 48.5  | 44.5 |
|                                                                                  |             | MCWB   | 35.0  | 36.0  | 38.1  | 43.5  | 51.5  | 53.8  | 60.2  | 57.1  | 53.4  | 45.8 | 41.6  | 37.8 |
| Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 41.7  | 43.9  | 45.2  | 51.7  | 60.0  | 61.1  | 68.1  | 63.4  | 60.6  | 54.6 | 49.5  | 44.8 |
|                                                                                  |             | MCDB   | 49.0  | 53.6  | 52.8  | 63.6  | 74.6  | 70.8  | 80.4  | 71.4  | 67.0  | 61.5 | 60.0  | 51.7 |
|                                                                                  | 2%          | WB     | 39.4  | 40.4  | 42.8  | 48.6  | 57.1  | 58.2  | 64.8  | 61.3  | 57.6  | 51.3 | 47.0  | 42.2 |
|                                                                                  |             | MCDB   | 46.3  | 47.7  | 48.9  | 59.6  | 68.2  | 66.2  | 74.5  | 68.7  | 64.2  | 59.3 | 54.7  | 48.3 |
|                                                                                  | 5%          | WB     | 37.4  | 38.4  | 40.8  | 46.8  | 54.6  | 56.6  | 63.0  | 60.0  | 55.8  | 49.0 | 44.7  | 40.2 |
|                                                                                  |             | MCDB   | 43.7  | 45.0  | 46.9  | 56.6  | 65.4  | 64.6  | 72.1  | 66.6  | 62.5  | 56.2 | 51.8  | 46.0 |
|                                                                                  | 10%         | WB     | 35.6  | 36.7  | 38.9  | 44.6  | 52.4  | 54.9  | 60.9  | 58.5  | 54.3  | 46.8 | 42.0  | 38.1 |
|                                                                                  |             | MCDB   | 40.1  | 42.4  | 44.0  | 52.7  | 62.7  | 62.5  | 68.5  | 64.8  | 60.4  | 53.4 | 47.2  | 43.7 |
| Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR   | 8.3   | 8.7   | 11.3  | 14.8  | 16.2  | 15.0  | 14.8  | 14.1  | 12.7  | 10.1 | 8.4   | 8.6  |
|                                                                                  |             | MCDBR  | 11.9  | 11.9  | 14.5  | 19.5  | 23.0  | 21.3  | 22.3  | 18.5  | 15.5  | 13.5 | 12.7  | 13.4 |
|                                                                                  |             | MCWBR  | 7.3   | 6.8   | 8.3   | 9.5   | 10.1  | 10.1  | 10.0  | 8.0   | 7.5   | 6.9  | 7.3   | 8.1  |
|                                                                                  | 5% WB       | MCDBR  | 11.7  | 12.0  | 14.1  | 18.4  | 21.2  | 18.3  | 20.7  | 16.0  | 15.3  | 12.7 | 11.7  | 13.1 |
| MCWBR                                                                            |             | 7.6    | 7.2   | 8.9   | 9.4   | 10.2  | 9.6   | 10.0  | 8.3   | 8.4   | 7.1   | 7.3  | 8.7   |      |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.244  | 0.302 | 0.333 | 0.360 | 0.360 | 0.362 | 0.371 | 0.358 | 0.337 | 0.312 | N/A  | 0.365 |      |
|                                                                                  | taud        | 2.031  | 2.185 | 2.225 | 2.282 | 2.362 | 2.424 | 2.433 | 2.468 | 2.504 | 2.396 | N/A  | 2.541 |      |
|                                                                                  | Ebn at Noon | 115    | 197   | 239   | 259   | 271   | 272   | 266   | 260   | 243   | 198   | N/A  | 61    |      |
|                                                                                  | Edh at Noon | 10     | 20    | 28    | 33    | 33    | 32    | 31    | 28    | 22    | 16    | N/A  | 6     |      |
| All-Sky Solar<br>Radiation                                                       | RadAvg      | 50     | 218   | 629   | 1178  | 1522  | 1589  | 1471  | 1134  | 683   | 302   | 79   | 25    |      |
|                                                                                  | RadStd      | 8      | 36    | 82    | 84    | 83    | 135   | 128   | 111   | 76    | 40    | 14   | 2     |      |

## Historical Trends

|      |                         | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|------|-------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|      |                         |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
| (a)  | (b)                     | (c)   | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)   | (k)   | (l)   |
| (46) | Station Only            | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| (47) | Regional (37 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | -257        | -340  | +121  | N/A   |

Nomenclature: See separate page